

$$\#1. \vec{r} = \vec{R} + \vec{\rho} = (R+x)\hat{i} + y\hat{j} + z\hat{k}$$

Conlon theorem,

$$\begin{aligned} \ddot{\vec{r}} &= (\ddot{R} + \dot{x})\hat{i} + \ddot{y}\hat{j} + \ddot{z}\hat{k} + (R+x)\dot{\omega}\hat{i} + y\dot{\omega}\hat{j} + z\dot{\omega}\hat{k} \\ &= \ddot{x}\hat{i} + \ddot{y}\hat{j} + \ddot{z}\hat{k} + \omega\hat{k} \times [(R+x)\hat{i} + y\hat{j} + z\hat{k}] \\ &= (\ddot{x} - \omega y)\hat{i} + (\ddot{y} + \omega(R+x))\hat{j} + \ddot{z}\hat{k} \end{aligned}$$

$$\begin{aligned} \ddot{\vec{r}} &= (\ddot{x} - \omega y)\hat{i} + (\ddot{y} + \omega(R+x))\hat{j} + \ddot{z}\hat{k} + \omega\hat{k} \times [(\ddot{x} - \omega y)\hat{i} + (\ddot{y} + \omega(R+x))\hat{j} + \ddot{z}\hat{k}] \\ &= \ddot{x}\hat{i} - \omega y\hat{j} - \omega y\hat{j} - \omega^2(R+x)\hat{i} + \ddot{y}\hat{j} + \omega(R+x)\hat{i} + \omega(R+x)\hat{j} - \omega^2 y\hat{j} + \ddot{z}\hat{k} \\ &= \ddot{x}\hat{i} - 2\omega y\hat{j} - \omega^2(R+x)\hat{i} + \ddot{y}\hat{j} + 2\omega(R+x)\hat{j} - \omega^2 y\hat{j} + \ddot{z}\hat{k} \end{aligned}$$

By Newton's Law,

$$\hat{i}: \ddot{x} - 2\omega y - \omega^2(R+x) = -\frac{\mu}{r^3}(R+x) + U_x$$

$$\hat{j}: \ddot{y} + 2\omega x - \omega^2 y = -\frac{\mu}{r^3}y + U_y$$

$$\hat{k}: \ddot{z} = -\frac{\mu}{r^3}z + U_z$$

$$\text{where } r = \sqrt{(R+x)^2 + y^2 + z^2}$$

b. For circular orbits, $\bar{x} = \bar{y} = \bar{z} = 0$.

Define $x = \bar{x} + \Delta x = \Delta x$, $y = \bar{y} + \Delta y = \Delta y$, $z = \bar{z} + \Delta z = \Delta z$.

$$\begin{aligned} \Rightarrow \frac{1}{r^3} &= \frac{1}{(\sqrt{(R+\Delta x)^2 + \Delta y^2 + \Delta z^2})^3} = \frac{1}{(R^2 + 2\Delta x R)^{3/2}} = \frac{1}{R^3} \left(1 + 2\frac{\Delta x}{R}\right)^{-3/2} \\ &= \frac{1}{R^3} \left(1 - \frac{3\Delta x}{R}\right) \end{aligned}$$

$$\hat{i}: \ddot{\Delta x} - 2\omega \Delta y - \omega^2 R - \omega^2 \Delta x = -\frac{\mu}{R^3} \left(1 - \frac{3\Delta x}{R}\right) (R + \Delta x) + U_x$$

$$= -\frac{\mu}{R^3} (R - 2\Delta x) + U_x$$

$$= -\omega^2 (R - 2\Delta x) + U_x$$

$$= -\omega^2 R + 2\omega^2 \Delta x + U_x$$

$$\Rightarrow \ddot{\Delta x} - 2\omega \Delta y - 3\omega^2 \Delta x = U_x$$

$$\hat{j}: \ddot{\Delta y} + 2\omega \Delta x - \omega^2 \Delta y = -\frac{\mu}{R^3} \Delta y \left(1 - \frac{3\Delta x}{R}\right) + U_y$$

$$= -\omega^2 \Delta y + U_y$$

$$\Rightarrow \ddot{\Delta y} + 2\omega \Delta x = U_y$$

$$\hat{k}: \ddot{\Delta z} = -\frac{\mu}{R^3} \Delta z \left(1 - \frac{3\Delta x}{R}\right) + U_z = -\omega^2 \Delta z + U_z$$

$$\Rightarrow \ddot{\Delta z} + \omega^2 \Delta z = U_z$$

$$\text{The Clohessy-Wiltshire equation} \begin{cases} \ddot{\Delta x} - 2\omega \Delta y - 3\omega^2 \Delta x = U_x \\ \ddot{\Delta y} + 2\omega \Delta x = U_y \\ \ddot{\Delta z} + \omega^2 \Delta z = U_z \end{cases}$$

where $\Delta x, \Delta y, \Delta z$ replaced x, y, z .

#2 a) Consider CW equation. The system matrix A is,

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ \omega^2 & 0 & 0 & +2\omega & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2\omega & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\omega^2 & 0 \end{bmatrix} \quad \text{where state } X = \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix}$$

If there is no external force, the equations of motion is $\dot{X} = AX$
 $\Rightarrow X(t) = \exp(At) X_0$ where X_0 is an initial state.

The state transition matrix $\Phi(t) = \exp(At)$ has nonlinear term for each component.

Hence the solution of CW eq. is nonlinear equation, and nonlinear terms are $\sin \omega t$ and $\cos \omega t$.

Now consider the approximation equations.

If there is no external force, $\ddot{x} = \ddot{y} = \ddot{z} = 0$.

Thus the solutions of the approximation eq are.

$$x_a(t) = \dot{x}_0 t + x_0, \quad y_a(t) = \dot{y}_0 t + y_0, \quad z_a(t) = \dot{z}_0 t + z_0$$

where $x_0, y_0, z_0, \dot{x}_0, \dot{y}_0, \dot{z}_0$ are the initial state.

Those solutions are linear approximation of the original solutions.

Hence, for any time $t < T$,

$$\left| \frac{x_a(t) - x(t)}{x(t)} \right| \leq 0.03 \quad \& \quad \left| \frac{y_a(t) - y(t)}{y(t)} \right| \leq 0.03 \quad \& \quad \left| \frac{z_a(t) - z(t)}{z(t)} \right| \leq 0.03$$

Since $\sin(\omega t)$ and $\cos(\omega t)$ are only nonlinear terms of x, y, z , then $\sin(\omega t)$ and $\cos(\omega t)$ define T .

$$\frac{\omega t - \sin(\omega t)}{\sin(\omega t)} \leq 0.03 \quad \& \quad \frac{1 - \cos(\omega t)}{\cos(\omega t)} \leq 0.03$$

$$\Rightarrow \omega t - 1.03 \sin \omega t \leq 0 \quad \& \quad 1 \leq 1.03 \cos(\omega t)$$

$$\Rightarrow \omega t \leq 0.2419$$

Since $0.2419 - 1.03 \sin(0.2419) = -4.8342 \times 10^{-3}$, then 0.2419 is the largest ωt satisfying both inequality equations.

Hence $T = 0.2419 / \omega = 0.2419 \sqrt{R^3 / \mu} \approx 204.4125 \text{ s} \approx 3.407 \text{ min}$.

Since the periods of $\sin(\omega t)$ and $\cos(\omega t)$ are $2\pi / \omega$.

Then t should be smaller than $0.03(2\pi / \omega) \approx 159.2842 \text{ sec}$.

Therefore, the maneuver time period $T = 159.2842 \text{ sec}$

b) Since all x, y, z are decoupled, then we can consider only x .

$$x(0) = -10, \quad x(60) = 0, \quad \dot{x}(0) = \dot{x}(60) = 0$$

$$\text{Since } x(t) = \dot{x}_0 t + x_0 = \dot{x}_0 t - 10, \text{ then } x(60) = 60\dot{x}_0 - 10 = 0$$

$$\Rightarrow \dot{x}_0 = 0.1667 \text{ km/s}$$

$$\text{Since } \dot{x}(t) = \dot{x}_0, \quad \dot{x}_f = \dot{x}_0 = 0.1667 \text{ km/s} \quad \& \quad \Delta \vec{V}_i = 0.1667 \hat{i} \text{ km/s}$$

$$\Delta \vec{V}_f = -0.1667 \hat{i} \text{ km/s}$$

#3 If (x_e, y_e, z_e) is an equilibrium point, then all equations of motion for the M-E circular restricted 3 body problem hold for (x_e, y_e, z_e) .
 with $\dot{x}_e = \dot{y}_e = \dot{z}_e = \ddot{x}_e = \ddot{y}_e = \ddot{z}_e = 0$.

$$\textcircled{2} -\omega_0^2 x_e = -\frac{\mu_E(x_e - D_E)}{|x_e - D_E|^3} - \frac{\mu_M(x_e + D_M)}{|x_e + D_M|^3} \dots (A)$$

$$\text{Left side: } -\omega_0^2 x_e = (2.66199 \times 10^{-6})^2 (-444645.6) \approx 3.1509 \times 10^{-6}$$

$$\begin{aligned} \text{Right side: } -\frac{\mu_E(x_e - D_E)}{|x_e - D_E|^3} &= -\frac{398601(-444645.6 - 4674)}{(444645.6 + 4674)^3} \\ &= \frac{398601}{(444645.6 + 4674)^2} \\ &\approx 1.9743 \times 10^{-6} \\ -\frac{\mu_M(x_e + D_M)}{|x_e + D_M|^3} &= -\frac{4887(-444645.6 + 380073)}{(444645.6 - 380073)^3} \\ &= \frac{4887}{(444645.6 - 380073)^2} \\ &\approx 1.1717 \times 10^{-6} \end{aligned}$$

$$\Rightarrow \text{Total } 3.1460 \times 10^{-6}$$

Since $3.1509 \times 10^{-6} \approx 3.1460 \times 10^{-6}$, then (A) holds for (x_e, y_e, z_e) .
 The other two equations hold trivially because $y_e = z_e = 0$.
 (Both sides are zero)

Hence (x_e, y_e, z_e) is an equilibrium point.

b. Let $x = x_e + \Delta x$, $y = y_e + \Delta y$, $z = z_e + \Delta z$.

$$r_E^{-3} = [(x_e + \Delta x - D_E)^2 + \Delta y^2 + \Delta z^2]^{-3/2} \approx [(x_e - D_E)^2 + 2\Delta x(x_e - D_E)]^{-3/2} \quad \text{HOT}$$

$$= (D_E - x_e)^{-3} \left[1 + 2\Delta x \frac{1}{x_e - D_E} \right]^{-3/2}$$

$$\approx (D_E - x_e)^{-3} \left[1 - 3\Delta x / (x_e - D_E) \right] \quad \text{Binomial thm}$$

$$r_M^{-3} = [(x_e + \Delta x + D_M)^2 + \Delta y^2 + \Delta z^2]^{-3/2}$$

$$\approx [(x_e + D_M)^2 + 2\Delta x(x_e + D_M)]^{-3/2}$$

$$= -(x_e + D_M)^{-3} \left[1 + 2\Delta x / (x_e + D_M) \right]^{-3/2}$$

$$\approx -(x_e + D_M)^{-3} \left[1 - 3\Delta x / (x_e + D_M) \right]$$

$$\textcircled{1} (\ddot{x}_e + \ddot{\Delta x}) + 2\omega_0(\dot{y}_e + \dot{\Delta y}) - \omega_0^2(x_e + \Delta x) = \frac{\mu_M}{|x_e + D_M|^3} (x_e + \Delta x) \left(1 - \frac{3\Delta x}{x_e + D_M} \right) + \frac{F_x}{m}$$

$$= -\frac{\mu_E}{(D_E - x_e)^3} (x_e - D_E) \left(1 - \frac{3\Delta x}{x_e - D_E} \right) + \frac{\mu_M}{(x_e + D_M)^3} (x_e + D_M) \left(1 - \frac{3\Delta x}{x_e + D_M} \right) + \frac{F_x}{m}$$

$$= -\frac{\mu_E}{(D_E - x_e)^2} (x_e - D_E) + 2\frac{\mu_E}{(D_E - x_e)^3} \Delta x + \frac{\mu_M}{(x_e + D_M)^2} (x_e + D_M) - 2\frac{\mu_M}{(x_e + D_M)^3} \Delta x + \frac{F_x}{m}$$

$$\Rightarrow \ddot{\Delta x} + 2\omega_0 \dot{\Delta y} - \omega_0^2 \Delta x = 2\frac{\mu_E}{(D_E - x_e)^3} \Delta x - 2\frac{\mu_M}{(x_e + D_M)^3} \Delta x + \frac{F_x}{m}$$

$$\Rightarrow \ddot{\Delta x} + 2\omega_0 \dot{\Delta y} - \omega_0^2 \left(1 + 2\frac{\mu_E}{\omega_0^2} - 2\frac{\mu_M}{\omega_0^2} \right) \Delta x = \frac{F_x}{m}$$

$$W_E^2 = \frac{\mu_E}{(D_E - x_e)^3} = \frac{398601}{(4674 + 444654.6)^3} \approx 0.6201 W_0^2$$

$$W_M^2 = \frac{\mu_M}{(x_e + D_M)^3} = -\frac{4887}{(444654.6 - 3800713)^3} \approx -2.5603 W_0^2$$

$$\therefore \ddot{\Delta x} - 2W_0 \dot{\Delta y} - 17.3608 W_0^2 \Delta x = \frac{F_x}{m}$$

$$\textcircled{1} \ddot{\Delta y} + 2W_0 \dot{\Delta x} - W_0^2 \Delta y = -W_E^2 \Delta y \left(1 + \frac{3\Delta x}{D_E - x_e}\right) + W_M^2 \Delta y \left(1 - \frac{3\Delta x}{x_e + D_M}\right) + \frac{F_y}{m}$$

$$= -W_E^2 \Delta y + W_M^2 \Delta y + F_y/m$$

$$\Rightarrow \ddot{\Delta y} + 2W_0 \dot{\Delta x} - (W_0^2 - W_E^2 + W_M^2) \Delta y = F_y/m$$

$$\therefore \ddot{\Delta y} + 2W_0 \dot{\Delta x} + 2.1804 W_0^2 \Delta y = \frac{F_y}{m}$$

$$\textcircled{2} \ddot{\Delta z} = -W_E^2 \Delta z \left(1 + \frac{3\Delta x}{D_E - x_e}\right) + W_M^2 \Delta z \left(1 - \frac{3\Delta x}{x_e + D_M}\right) + \frac{F_z}{m}$$

$$= -W_E^2 \Delta z + W_M^2 \Delta z + F_z/m$$

$$\Rightarrow \ddot{\Delta z} + (W_E^2 - W_M^2) \Delta z = F_z/m$$

$$\therefore \ddot{\Delta z} + 3.1804 W_0^2 \Delta z = \frac{F_z}{m}$$

Replace $\Delta x, \Delta y, \Delta z$ to x, y, z .

$$\begin{cases} \ddot{x} - 2W_0 \dot{y} - 17.3608 W_0^2 x = F_x/m \\ \ddot{y} + 2W_0 \dot{x} + 2.1804 W_0^2 y = F_y/m \\ \ddot{z} + 3.1804 W_0^2 z = F_z/m \end{cases}$$

$$\ddot{y} + 2W_0 \dot{x} + 2.1804 W_0^2 y = F_y/m$$

$$\ddot{z} + 3.1804 W_0^2 z = F_z/m$$

#4. a. Large, Periodic orbit such that is a solution of original nonlinear differential equations of motions

b. Rendezvous

- State to state transfer

position and velocity.

6 boundary conditions at terminal time.

- Orbit transfer

State to reduced state transfer.

5 boundary conditions at terminal time.

An angular displacement at terminal time is not considered.

- Interception

State to position transfer

3 boundary conditions at terminal time.

A velocity at terminal time is not considered.